KESTREL · RESEARCH PRESENTATION

Calibrated cascade routing for on-device intent understanding

A pre-registered evaluation on a laptop CPU

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Mentor review · October 2026 · draft

THE PROBLEM

A local language model costs almost a second

Tier 4 · Gemma-4-E2B, 4-bit

865.0 ms

median per request
3491 MiB resident

Tier 3 · e5-small encoder + linear head

5.4 ms

median per request
278 MiB resident

Most requests are simple. Can cheap tiers answer them, and send only the rest to the model, without losing accuracy?

Measured on CLINC150 test, Ryzen 5 5500U, 6 threads.

Answer at the cheapest confident tier

Multi-intent signal

Does the request contain *and*, *then*, *aur*, *phir* or *fir*?

yes: straight to Tier 4

→

Tier 1 · rules

Keyword rules for the Kestrel intents

confidence = precision on calibration data

≈ microseconds

accept if $p \geq \tau_1$

→

Tier 3 · encoder

multilingual e5-small (Q8_0) + linear head

temperature T_3 fitted by NLL

≈ 5 ms

accept if $p \geq \tau_3$

→

Tier 4 · SLM

Gemma-4-E2B (Q4_0), labels only

confidence: label probability, T_4 by BCE

≈ 0.9 s

accept if $p \geq \tau_4$, else abstain

No tier confident → abstain: ask the user to clarify, or refuse.

Every τ and T comes from calibration and validation data, frozen before test.

On CLINC150 the cascade is Tier 3 → Tier 4: the Kestrel rules cannot answer with its labels.

What the closest prior work measures

Work	What is cascaded	Calibrated	Cost measured	Rule tier	Code-mixed
C3 (Lu 2024)	encoders or LLMs, by size	yes	FLOPs	no	no
CascadeBERT (Li 2021)	BERTs, by depth	yes	FLOPs	no	no
CAUC (Zhang 2026)	2–7 models, same labels	yes	parameters, GFLOPs	no	no
Arora 2024	encoder → cloud LLM	no	API latency	no	no
Chaganti 2026	NER encoders, by size	yes	latency on device	no	no
Sadhu 2026	router → solver / SLM	n/a	latency, energy (Pi)	yes	no
This work	**rules → encoder → SLM**	**yes**	**latency, RAM (laptop)**	**yes**	**Hinglish**

No paper in our search matches every column, but each column has close prior work. So this is an evaluation, not a new method.

53 references checked against primary sources, none unverified.

Pre-registered hypotheses

Question	Hypothesis	Data
RQ1 · H1 (primary)	**Cascade R3 within $\delta = 0.0263$ of always-SLM R0 in accuracy, and faster**	**CLINC150**
RQ1 · H1b	R3 more accurate than random escalation at the same cost	both
RQ1 · H1c	R3 within δ of the encoder alone (R1)	both
RQ2 · H2	Calibration lowers selective risk at 80% coverage	both
RQ3 · H3	Adding the encoder to rules + SLM: accuracy within δ, lower cost	Intents-v0
RQ5 · H5	Replay forgets less than naive updates of the head	CLINC150
RQ7 · H7a, H7b	Hinglish: R3 within δ of R0; higher calibration error than English	Intents-v0
RQ4, RQ6	H4 personalization, H6 quantization: pending the dataset review	—

H1 is tested alone. The 13 secondary members are adjusted together by Holm's method.

METHOD

How it was tested

Frozen before test

- Pre-registration frozen as a git tag
- $\delta = 0.0263$ from a 5-seed pilot on validation
- Thresholds and temperatures from calibration and validation only
- Holm across 13 secondary hypotheses

Fair baselines

- R0: always the SLM, tuned abstention threshold
- R1: the encoder alone, tuned the same way
- R-rand: random escalation at R3's rates, same cost
- Same models, prompts, grammar and threads

Measured on the laptop

- Ryzen 5 5500U, 6 cores pinned, turbo off
- Median of 5 timed repetitions after warm-up
- 5 invalidation rules; every test run valid first time
- Peak RSS (VmHWM), models loaded without mmap

CLINC150 test: 5500 requests, 1 000 out of scope · Kestrel-Intents-v0 test: 252 synthetic requests, English + Hinglish

Paired bootstrap, 10 000 resamples; Intents-v0 resamples its 44 template families.

PRIMARY HYPOTHESIS H1 · CLINC150 TEST

Supported: as accurate as always-SLM, and faster

+0.048

accuracy, R3 − R0 · 95% CI [+0.038, +0.058]

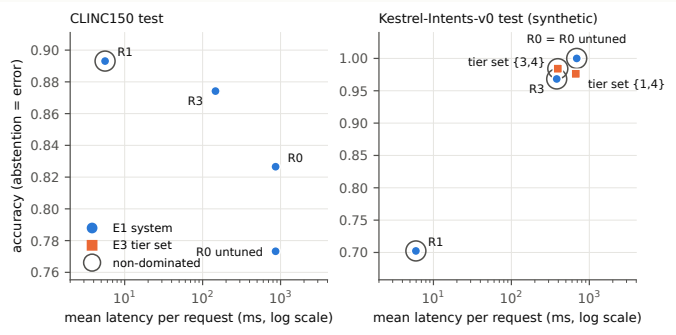
−717 ms

mean latency per request · 95% CI [−739, −694]

On 5500 requests: R3 0.874 vs tuned R0 0.827 accuracy; 17.0% reached the SLM, which answered 6.0%.

Margin $\delta = 0.0263$ · dropping the 14 duplicated test items leaves the CI unchanged.

On CLINC150 the encoder alone was better still



CLINC150

R1 0.893 at 6 ms beats R3 0.874 at 146 ms: more accurate and faster

Intents-v0

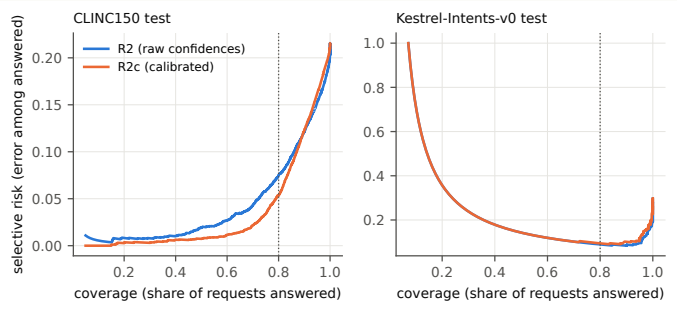
R0 is perfect (1.000 at 689 ms); R3 0.968 at 382 ms; R1 only 0.702

Memory

The cascade saves time, not RAM: 3773 vs 3491 MiB peak

Ringed points are not dominated · H1c on CLINC150 inconclusive (Holm p 0.27).

Calibration lowered risk on CLINC150 only



CLINC150 · supported

Risk at 80% coverage 0.0748 → 0.0539; Δ −0.0209 [−0.0263, −0.0148]

Encoder ECE 0.0553 → 0.0083

Intents-v0 · inconclusive

0.0891 → 0.0941; Δ +0.0050 [−0.0007, +0.0199]

Why it helps

Per-tier risk barely moved; the gain is making tiers comparable under one threshold

R2: raw confidences · R2c: temperature-scaled · one shared threshold.

The encoder tier pays; replay stops forgetting

E3 tier ablation · Intents-v0 · H3 supported

−289 ms

mean cost of adding the encoder to rules + SLM · CI $[-371, -211]$

Accuracy Δ -0.008 $[-0.020, +0.000]$, within δ

Rules cost accuracy: {1,4} 0.976 vs {4} 1.000

E5 head updates · CLINC150, 15 runs · H5 supported

0.918

average accuracy with replay, vs 0.188 naive (joint 0.928)

Forgetting: naive 0.947, replay -0.002; EWC $\approx$ naive (0.932)

Price: a buffer of half of each old class's training items, as embeddings (9.3 MB)

Tier 2 (distilled model) was not built · E5: 10 domains in 5 stages, 3 orders × 5 seeds.

ALL HYPOTHESES

Every pre-registered test, with its outcome

1 of 1 primary supported

6 of 13 secondary supported

5 inconclusive

0 not supported

2 pending (E4, E6)

Hypothesis	Holm p	Outcome
H1 CLINC150 (primary)	$< 10^{-4}$ (alone)	supported
H1b CLINC150	$< 10^{-4}$	supported
H1b Intents-v0	0.002	supported
H1c CLINC150	0.27	inconclusive (interim)
H1c Intents-v0	$< 10^{-4}$	supported
H1 Intents-v0	1	inconclusive (interim)
H2 CLINC150	$< 10^{-4}$	supported

Hypothesis	Holm p	Outcome
H2 Intents-v0	1	inconclusive (interim)
H3 encoder tier	0.045	supported
H5 replay vs naive	6.1×10^{-4}	supported
H7a Hinglish accuracy	0.25	inconclusive (interim)
H7b Hinglish ECE	0.30	inconclusive (interim)
H4 personalization	—	pending
H6 quantization	—	pending

H1 tested alone · Holm interim until H4, H6 run: “supported” is final, “inconclusive” may change.

Why the cascade lost where it did

CLINC150 · out-of-scope shift

68%

of R3's 692 errors are out-of-scope requests answered as in-scope

Out-of-scope share: 11.8% of validation, 18.2% of test

149 out-of-scope test requests fall between R3's 0.601 and R1's 0.782 thresholds

Intents-v0 · multi-intent gap

6 of 8

of R3's errors are two-part requests joined by "also" or ", now"

The signal knows only *and, then, aur, phir, fir*; all 24 validation compounds had one

Fixes (Tier-3 reject band, out-of-scope detector, wider signal) need a new pre-registered version.

ECE 0.0083 after scaling covers in-scope items only; calibration on out-of-scope requests is untested.

LIMITATIONS

What these results do not show

One machine
one CPU, one SLM, one encoder; turbo off

Synthetic data
Intents-v0 drafted with AI help; κ not yet measured

Small Hinglish set
126 requests per language; wide intervals

E4, E6 pending
personalization and quantization wait on the dataset review

Warm process only
latency excludes model loading; one warm, single-user process

Reproduced here only
byte-identical reruns here; a second machine next week

δ before SESOI
margin frozen before the smallest effect of interest was confirmed

No energy
the only counter (RAPL) is root-only here

Privacy tested
11 of 11 checks pass; not a formal guarantee

Full list: experiments/LIMITATIONS.md and the paper's threats to validity.

What the evidence supports, and what is next

Answers

RQ1 · yes against the SLM on CLINC150 ($+0.048$, -717 ms); no against the encoder alone

RQ2 · calibration lowers risk on CLINC150; not replicated on synthetic data

RQ3 · the encoder tier changes mean cost by -289 ms, accuracy within δ

RQ5 · replay prevents forgetting; EWC does not

RQ7 · Hinglish: inconclusive

Next

1. Review the Tasks and Personas datasets; freeze amendment v1.1
2. Run E4 and E6; final Holm family
3. Reproduce on a second machine (Mac)
4. Second annotator (κ); confirm the smallest effect of interest

Paper draft, code, logs and the one-command reproduction (experiments/repro.py) are in the repository.